

Black-Box Watermark Removal Using Diffusion Models and Self-Attention Mechanisms

1st Qingyuan Zeng*Institute of Artificial Intelligence**Xiamen University*

Xiamen, China

36920221153145@stu.xmu.edu.cn

2nd Yunpeng Gong*School of Informatics**Xiamen University*

Xiamen, China

fmonkey625@gmail.com

3rd Chuangliang Zhang*School of Informatics**Xiamen University*

Xiamen, China

31520231154325@stu.xmu.edu.cn

4th Shu Jiang*Institute of Artificial Intelligence**Xiamen University*

Xiamen, China

36920241153222@stu.xmu.edu.cn

5th Zhenzhong Wang*School of Informatics**Xiamen University*

Xiamen, China

zhenzhongwang@xmu.edu.cn

6th Min Jiang**School of Informatics**Xiamen University*

Fujian, China

minjiang@xmu.edu.cn

Abstract—Digital watermarking has become a cornerstone for copyright protection in the digital era, yet its robustness against removal attacks remains a critical challenge. Conventional watermark removal techniques often degrade image quality or fail to counter advanced watermarking methods. This paper introduces a novel Diffusion-Based Watermark Removal Attack (DWRA), harnessing the generative power of diffusion models to effectively erase watermarks while preserving image fidelity. Operating in a black-box setting, DWRA requires no prior knowledge of watermark embedding or detection mechanisms. By projecting watermarked images into a deep feature space, applying noise to disrupt watermark features, and employing a self-attention diffusion model for denoising, our approach achieves watermark removal with minimal quality loss. Extensive experiments across diverse watermarking techniques demonstrate that DWRA surpasses existing methods in both removal efficacy and image quality preservation, exposing vulnerabilities in current watermarking systems and emphasizing the need for more resilient designs.

Index Terms—Watermark removal, black-box attack, self-attention, diffusion models.

I. INTRODUCTION

Digital watermarking is a cornerstone of media copyright protection, enabling content creators to embed imperceptible identifiers into images and videos. With the proliferation of AI-generated content, these techniques have gained renewed importance for tracing provenance and preventing misuse [1], [2]. Modern watermarking methods, such as frequency-domain hybrids (DwtDct, DwtDctSvd) [3] and deep learning-based approaches (RivaGAN, Stable Signature) [4], achieve remarkable robustness against conventional distortions like JPEG compression, brightness adjustment, and Gaussian noise. Yet, their resilience against adversarial removal attacks—deliberate attempts to erase watermarks—remains underexplored.

The corresponding author: Min Jiang (minjiang@xmu.edu.cn). This work was supported in part by the National Natural Science Foundation of China under Grant No. 62276222.

Existing removal strategies suffer from critical limitations. Traditional image processing attacks (e.g., contrast adjustment [5], blurring [6]) degrade visual quality excessively and fail against learning-based watermarks [4]. Recent deep learning attacks [7], [8] leverage surrogate models and gradient-based optimization but assume white-box access to the watermark detector or embedding parameters—an unrealistic scenario for real-world adversaries. Crucially, no existing method efficiently removes watermarks under black-box conditions (i.e., no detector/embedder access) while preserving image quality [9]–[15]. This gap leaves watermarking systems exposed to practical threats and hinders objective robustness evaluation.

To bridge this gap, we introduce DWRA, a black-box attack that combines latent-space noise disruption with diffusion-based reconstruction. Our key insight is that watermarks manifest as localizable high-frequency artifacts in a well-structured feature space [16], [17]. By projecting watermarked images into this space (via VAE), perturbing watermark-associated features with noise, and reconstructing the image using a self-attention diffusion model conditioned on global semantics, DWRA decouples watermark removal from quality degradation. The core innovation is SUNet—a self-attention-enhanced U-Net diffusion model—that leverages spatial dependencies to restore coherent image structures after noise injection (Fig. 2). This allows DWRA to operate without watermarking system details while outperforming existing attacks.

- 1) We define the black-box watermark removal task, where attackers erase watermarks without detector/embedder access while preserving image quality.
- 2) We propose DWRA, a framework combining latent-space noise injection and self-attention diffusion (SUNet) for watermark removal. SUNet’s global feature aggregation enables high-fidelity reconstruction post-attack.
- 3) Extensive experiments have fully demonstrated that our

DWRA method, in the context of black-box scenarios for watermark removal tasks, exhibits significantly superior performance compared to traditional methods, and also holds a distinct edge in preserving the integrity of image quality.

II. RELATED WORK

A. Invisible Watermarking Techniques

Modern watermarking embeds identifiers imperceptibly via spatial/frequency-domain perturbations or deep learning. Traditional methods like DWT-DCT and DWT-DCT-SVD [3] decompose images (e.g., using wavelets), modify transform coefficients (e.g., DCT, SVD), and reconstruct watermarked images. These achieve high quality but are vulnerable to geometric attacks [5], [6]. Deep learning-based methods improve robustness: RivaGAN [4] uses adversarial training with attention masks to embed watermarks resilient to cropping and compression. SSL-based techniques [18] employ self-supervised learning for detector-agnostic embedding, while Stable Signature [19] fine-tunes latent diffusion decoders. Though robust against traditional attacks, their resilience to learning-based removal—especially in black-box settings—is unproven.

B. Watermark Removal Attacks

Removal strategies fall into two categories: Traditional Image Processing and Learning-Based Removal.

1) *Traditional Image Processing*: Conventional watermark removal relies on signal-distorting operations that indiscriminately degrade image quality. Brightness and contrast adjustments [5] globally shift pixel distributions, reducing watermark detectability at the cost of unnatural visual artifacts. JPEG compression attacks [20] exploit quantization artifacts to disrupt high-frequency watermark signals but introduce blocking distortions. Gaussian noise injection [21] obscures watermarks by increasing background randomness, while Gaussian blurring [6] suppresses high-frequency components through spatial averaging—both sacrificing critical image details.

Geometric attacks like rotation or cropping misalign watermark patterns but cause visible structural distortions. Though computationally efficient, these methods fundamentally trade removal efficacy for perceptual fidelity and fail against modern adaptive watermarks. BM3D [22] treat watermarks as noise and apply advanced filters. These often oversmooth textures and edges, failing against watermarks integrated into semantically meaningful features.

2) *Learning-Based Removal*: These methods typically construct a surrogate model that mimics the behavior of the watermark detection system and utilize gradient-based optimization techniques to generate adversarial perturbations that can deceive the detector. For example, A study [7] have employed a substitute classifier trained on unlabeled data to approximate the decision boundary of the target watermark detector. By iteratively applying projected gradient descent (PGD) attacks on this surrogate model, they can produce

adversarial examples that are likely to be misclassified by the actual detector.

Another work [8] have proposed differentiable proxy keys to optimize the attack parameters in a manner that locally replicates the secret watermark key. This allows for a more targeted and efficient search for effective perturbations. Nevertheless, these learning-based methods often come with stringent requirements regarding the knowledge of the watermarking system. They typically assume a white-box setting where the attacker has access to the internal details of the watermark detector or the embedding parameters. In reality, however, attackers rarely possess such comprehensive information.

III. METHODOLOGY

A. Framework Overview

To address the challenge of removing digital watermarks from images without knowledge of the watermarking algorithms (a black-box scenario) while preserving image quality, this chapter introduces the Diffusion-based Watermark Removal Attack (DWRA) framework, as illustrated in Fig. 1. The DWRA framework operates by first mapping the watermarked image into a feature space to obtain watermarked features. Noise is then applied to disrupt the watermark representation within these features. To derive features free of watermarks, the DWRA method performs denoising on the noisy features. Finally, the framework maps these clean features back to the pixel space to generate a watermark-free image.

As depicted in Fig. 2, this study designs a U-Net model enhanced with a self-attention mechanism, termed SUNet. SUNet improves upon the traditional U-Net architecture by incorporating self-attention mechanisms in each layer of different resolutions within the encoder and decoder. This allows the capture of global features across different resolutions, enabling more refined image reconstruction. Since SUNet is trained on data without watermarks, it treats watermark features as part of the noise, effectively removing them to obtain clean features. The following sections detail the DWRA framework through training and attack phases.

B. Training Stage

The DWRA method utilizes a Variational Autoencoder (VAE) to establish a mapping between the pixel space and the latent feature space. The encoder $\mathbb{E}_{\text{VAE}}$ maps an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ to latent distribution parameters $(\boldsymbol{\mu}, \boldsymbol{\sigma})$, and samples the latent feature $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$ (where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$) using the reparameterization trick. The decoder $\mathbb{D}_{\text{VAE}}$ reconstructs the image from the latent feature as $\hat{\mathbf{x}} = \mathbb{D}_{\text{VAE}}(\mathbf{z})$. The training objective includes both reconstruction loss and KL divergence regularization, with the loss function defined as:

$$\mathbb{L}_{\text{VAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \frac{\beta}{2} (\boldsymbol{\mu}^T \boldsymbol{\mu} + \boldsymbol{\sigma}^T \boldsymbol{\sigma} - \log \boldsymbol{\sigma}^T \boldsymbol{\sigma} - 1) \quad (1)$$

Here, β is a balancing coefficient. It is important to note that this stage uses only a dataset of original images without watermarks $\{\mathbf{x}_i\}$, ensuring that the mapping from the latent

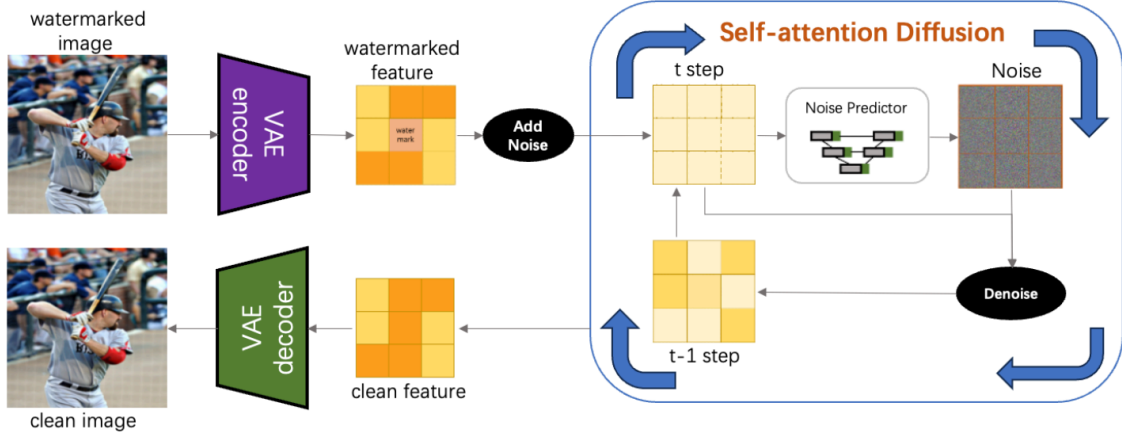


Fig. 1: Watermark removal framework DWRA.

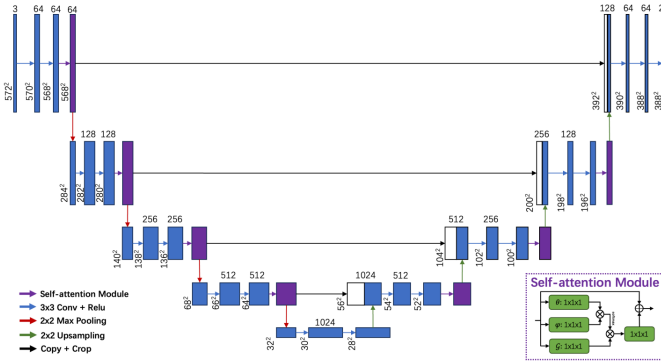


Fig. 2: Framework of the noise predictor SUNet.

feature space to the pixel space primarily captures the semantic features of the image rather than watermark information.

Subsequently, the training of the noise predictor SUNet focuses on enhancing the denoising capability within the latent feature space. Given the latent feature $\mathbf{z}_0 = \mathbb{E}_{\text{VAE}}(\mathbf{x})$, the diffusion process gradually adds Gaussian noise to generate noisy features $\mathbf{z}_t = \sqrt{\alpha_t}\mathbf{z}_0 + \sqrt{1-\alpha_t}\epsilon$ (where α_t is a predefined noise schedule parameter, and $t \sim \mathbb{U}(1, T)$ is a random time step). The SUNet model ϵ_θ , centered on the self-attention mechanism, comprises a downsampling encoder, an intermediate bottleneck layer, and an upsampling decoder, with each layer equipped with multi-head self-attention modules to capture long-range dependencies. The model takes as input the noisy feature $\mathbf{z}_t$ and the time step embedding $\mathbf{t}$, and outputs the predicted noise $\hat{\epsilon} = \epsilon_\theta(\mathbf{z}_t, t)$. The training objective employs the following mean squared error loss:

$$\mathbb{L}_{\text{diff}} = \mathbb{E}_{\mathbf{z}_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t)\|_2^2] \quad (2)$$

Parameters θ are optimized through gradient descent. Since the training data contains no watermark information, the model learns to suppress watermark features as part of the noise during the denoising process. The entire training procedure involves separate optimization in two stages, ensuring efficient

semantic encoding of the feature space and robust removal of anomalies (such as watermarks) by the model.

C. Attack Stage

During the attack phase, the DWRA framework removes watermarks robustly through noise injection in the feature space and iterative denoising using SUNet. Given a watermarked image $\mathbf{x}_w \in \mathbb{R}^{H \times W \times 3}$, the pre-trained VAE encoder first extracts the latent feature $\mathbf{z}_w = \mathbb{E}_{\text{VAE}}(\mathbf{x}_w) \in \mathbb{R}^{h \times w \times d}$, where $h = H/s$, $w = W/s$ (with s being the downsampling factor), and d is the dimension of the latent space. Next, controlled noise perturbations are applied in the latent space to construct noisy features:

$$\mathbf{z}_t = \sqrt{\alpha_t}\mathbf{z}_w + \sqrt{1-\alpha_t}\epsilon, \quad \epsilon \sim \mathbb{N}(0, \mathbf{I}) \quad (3)$$

Here, $\alpha_t \in (0, 1]$ is the noise scheduling coefficient (set using a cosine schedule $\alpha_t = \cos^2(\pi t/2T)$), and $t \in \{1, \dots, T\}$ is a randomly selected diffusion step. This process disrupts the local correlations of the latent features, scrambling the watermark information embedded in both the frequency and spatial domains. Subsequently, the noisy feature $\mathbf{z}_t$ is fed into the trained SUNet noise predictor ϵ_θ to estimate the injected noise:

$$\hat{\epsilon} = \epsilon_\theta(\mathbf{z}_t, t) \quad (4)$$

Through the reverse diffusion process, the denoised latent feature is reconstructed:

$$\mathbf{z}'_0 = \frac{1}{\sqrt{\alpha_t}} (\mathbf{z}_t - \sqrt{1-\alpha_t}\hat{\epsilon}) \quad (5)$$

Since SUNet is trained solely on features from original images without watermarks, its noise prediction process inherently identifies and suppresses watermark features as anomalies, thereby achieving removal of the watermark information. Finally, the purified feature is mapped back to the pixel space via the VAE decoder:

$$\mathbf{x}' = \mathbb{D}_{\text{VAE}}(\mathbf{z}'_0) \quad (6)$$

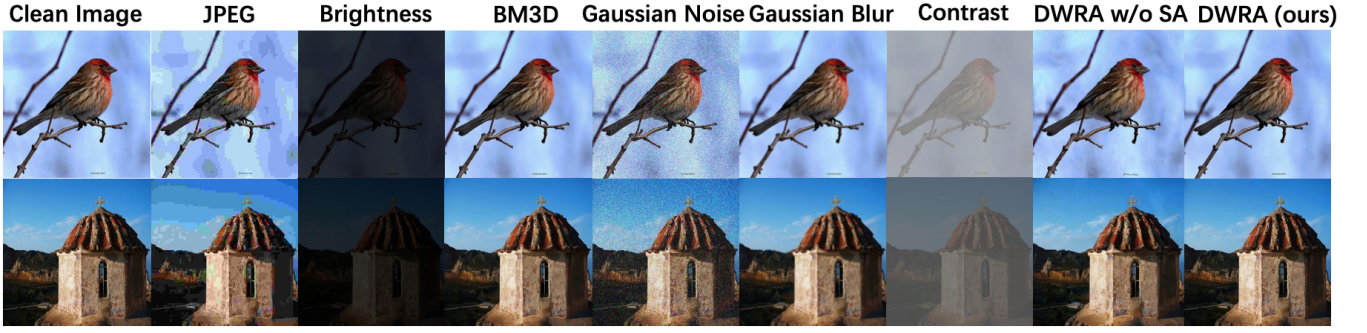


Fig. 3: Visual comparison of different image watermark removal methods.

The entire attack process relies entirely on the forward propagation of the VAE and SUNet. As their training requires no feedback from the watermark detector or prior knowledge of the watermark embedding parameters, this satisfies the requirements of a black-box watermark removal attack.

TABLE I: Similarity comparison between clean images and watermarked images with different watermark techniques

	SSIM $\uparrow$	PSNR (dB) $\uparrow$	MSE $\downarrow$	FID $\downarrow$
DwtDct	0.9996	60.6710	0.0584	2.0671
DwtDctSvd	0.9997	60.9608	0.0545	6.2081
RivaGAN	0.9970	48.4963	0.9216	6.0474

IV. EVALUATION AND RESULTS

A. Watermark Embedding Techniques

This section selects the most commonly used watermark embedding techniques, namely DWT-DCT, DWT-DCT-SVD, and RivaGAN methods, to embed the watermark byte codes mentioned in the previous section into images [3], [4]. Among these, DWT-DCT and DWT-DCT-SVD are traditional image processing-based watermark embedding techniques, while RivaGAN is a deep learning-based method.

B. Baseline Attack Algorithms

To verify the effectiveness of the proposed image watermark removal attack framework DWRA, this experiment selects a series of commonly used baseline attack methods as comparison objects for DWRA [5], [6], [20], [21]: JPEG compression attacks, brightness and contrast adjustment attacks, Gaussian noise attacks, Gaussian blur attacks, and BM3D attacks. Among these, JPEG compression attacks utilize the lossy nature of JPEG compression to degrade image quality and destroy watermark information. Brightness and contrast adjustment attacks interfere with watermarks by adjusting the brightness and contrast of images. Gaussian blur and Gaussian noise attacks increase the noise level of images, making watermark information less discernible during extraction. BM3D attacks leverage block matching and three-dimensional filtering for image denoising, indirectly removing watermarks by eliminating noise from images.

C. Comparison of Watermarked Image Quality

This experiment shows the image quality of three watermark embedding methods—DWT-DCT, DWT-DCT-SVD, and RivaGAN—using the ImageNet dataset. The results are shown in Table I. Four metrics are used: average SSIM, PSNR, MSE, and FID. Higher SSIM and PSNR indicate better quality, while lower MSE and FID suggest smaller differences from the original. The table shows DWT-DCT and DWT-DCT-SVD have comparable performance. Both achieve high SSIM (0.9996 and 0.9997), PSNR (60.6710 and 60.9608 dB), and low MSE (0.0584 and 0.0545), showing minimal visual impact. However, their FID values (2.0671 and 6.2081) indicate some distribution differences from the originals. RivaGAN has slightly lower SSIM (0.9970), PSNR (48.4963 dB), and higher MSE (0.9216), suggesting a minor visual quality decline. But its FID (6.0474) is close to DWT-DCT-SVD's, indicating high distribution similarity. Overall, all three methods generate watermarked images very similar to the originals.

D. Comparison of Image Quality in Watermark Removal

In this experiment, typical attacks like BM3D, JPEG compression, brightness, contrast, Gaussian blur and Gaussian noise are applied to simulate real-world interferences for various watermark embedding methods. DWRA and its DWRA w/o SA variant are also tested to assess SA's impact on watermark robustness.

Table I in appendix and table II in main paper show that under DwtDct, common attacks degrade image quality. For instance, brightness attacks lead to a low SSIM of 0.2457, PSNR of 7.6749 dB, and high MSE and FID. In contrast, DWRA achieves SSIM of 0.8884, PSNR of 32.4313 dB, MSE of 41.4785, and FID of 33.4569, surpassing DWRA w/o SA (SSIM 0.7391, PSNR 25.6370 dB). This proves SA enhances feature learning and adaptation in watermarked areas. Similar trends appear in DwtDctSvd and RivaGAN. In RivaGAN, DWRA gets SSIM of 0.8984 and PSNR of 33.3151 dB, while the SA - less version gets SSIM of 0.7490 and PSNR of 25.6298 dB. Overall, DWRA outperforms common attacks across all methods, effectively reducing MSE and FID under complex attacks like JPEG compression and Gaussian noise, ensuring high image similarity.

TABLE II: The attack performance of different watermark removal methods under various watermark embedding techniques.

Watermarking Method	Attack Method	Average SSIM $\uparrow$	Average PSNR (dB) $\uparrow$	Average MSE $\downarrow$	FID $\downarrow$
DwtDctSvd	BM3D	0.7917	28.8375	130.9428	38.5496
	JPEG	0.6901	15.0347	523.3080	82.5828
	Brightness	0.2457	7.6748	12163.9042	68.1190
	Contrast	0.5818	15.0966	2151.9985	59.4630
	GaussianBlur	0.6051	19.3637	561.7842	89.1373
	GaussianNoise	0.2718	18.2903	965.2206	105.1299
	Rotate	0.2226	9.8883	7244.4757	202.5867
	DWRA w/o SA	0.7378	25.6100	208.3521	43.3506
	DWRA	0.8881	32.4219	41.5756	34.5649
RivaGAN	BM3D	0.8016	29.0608	121.8186	38.1972
	JPEG	0.7386	18.8475	424.0828	80.9738
	Brightness	0.2441	7.6655	12189.7142	66.6618
	Contrast	0.5825	15.0945	2153.1599	59.7739
	GaussianBlur	0.6542	21.3068	562.1945	98.6057
	GaussianNoise	0.2724	18.2920	964.9844	104.3127
	Rotate	0.2216	9.8872	7246.0590	199.2101
	DWRA w/o SA	0.7490	25.6298	207.0995	45.8174
	DWRA	0.8984	33.3151	42.3506	35.2896

TABLE III: The watermark performance indicators of various attack methods under different watermarking technologies.

Watermarking	Attack Method	BitAcc $\downarrow$	WS $\downarrow$
DwtDctSvd	BM3DAttacker	0.7	0.351
	JPEG	0.771875	0.448
	Brightness	0.48125	0.0
	Contrast	0.475	0.0
	GaussianBlur	0.9984375	1.0
	GaussianNoise	0.503125	0.0
	Rotate	0.4734375	0.0
	DWRA w/o SA	0.4671875	0.064
	DWRA	0.4265625	0.042
RivaGAN	BM3DAttacker	0.84375	0.744
	JPEG	0.96875	1.0
	Brightness	0.575	0.0
	Contrast	0.5765625	0.0
	GaussianBlur	0.9984375	1.0
	GaussianNoise	0.7640625	0.465
	Rotate	0.596875	0.0
	DWRA w/o SA	0.5984375	0.0
	DWRA	0.55625	0.0

In summary, experiments show DWRA's superiority in resisting watermark removal attacks. The incorporated SA mechanism greatly boosts watermark robustness and imperceptibility, helping maintain good image quality in various attack scenarios.

E. Visual Comparison of Watermark Removal on Images

This experiment visually compares the impact of various watermark removal attack methods on image quality. As shown in Figure 3, existing attack methods, such as JPEG compression, brightness adjustment, BM3D, Gaussian noise, Gaussian blur, and contrast adjustment, all degrade image quality to varying extents. For example, JPEG compression introduces artifacts and blurriness, brightness attacks distort colors and details, BM3D attacks cause blurriness in texture-rich areas, Gaussian noise makes images chaotic, Gaussian

blur smoothes edges, and contrast attacks disrupt natural visual effects by making images overly dim or bright.

In contrast, DWRA method enhanced by the self-attention mechanism, significantly improves image quality. Images processed by DWRA closely resemble clean images, with well-preserved details, colors, and structure. This demonstrates that DWRA effectively minimizes quality degradation while removing watermarks, maintaining the visual integrity of the images.

F. Quantitative Comparison of Attack Performance

This experiment evaluates the attack performance of the proposed DWRA method and its variant (DWRA w/o SA) against common watermark removal methods across three watermark techniques: DwtDctSvd, and RivaGAN. The evaluation metrics are BitAcc (bit accuracy) and WS (watermark extraction success rate). Lower BitAcc and WS values indicate better watermark removal performance.

As shown in Table III, For DwtDctSvd, common attacks have BitAcc values from 0.47 to 0.7 and WS from 0 to 0.35, with inconsistent results. For example, Gaussian blur has high BitAcc (0.9984375) and WS (1.0), indicating ineffectiveness. In contrast, DWRA (BitAcc: 0.4265625) and DWRA w/o SA (BitAcc: 0.4671875) achieve WS of around 0.05, demonstrating stronger robustness. With RivaGAN, common attacks show significant variance. BM3D has BitAcc of 0.84375 and WS of 0.744, while JPEG compression has BitAcc of 0.96875 and WS of 1.0, indicating poor performance. DWRA (BitAcc: 0.55625) and DWRA w/o SA (BitAcc: 0.5984375) achieve WS of 0, proving effective in removing RivaGAN watermarks.

In summary, the DWRA method consistently outperforms common attacks across all three watermark techniques, effectively reducing watermark extraction accuracy and success rate. The self-attention mechanism in DWRA further enhances its performance, demonstrating strong applicability and robust attack capabilities.

TABLE IV: The average attack time of one single image with different attack methods.

Attack Method	Average time (Sec.) ↓
BM3D	17.2093
JPEG	0.0046
Brightness	0.0054
Contrast	0.0070
GaussianBlur	0.0043
GaussianNoise	0.0348
Rotate	0.0050
DWRA w/o SA	0.1964
DWRA (ours)	0.2073

G. Comparison of Attack Efficiency

This experiment evaluates the computational speed of various watermark removal attack methods to assess their efficiency in practical applications. Table IV shows the average processing time per image for each method.

Traditional image processing methods, such as brightness, contrast, Gaussian blur, Gaussian noise, rotation, scaling, cropping, and JPEG compression attacks, have average processing times ranging from 0.0043 to 0.0348 seconds per image. These methods are computationally efficient and can quickly complete attack tasks with minimal resource consumption. However, the BM3D is significantly slower, requiring an average of 17.2093 seconds per image. This is due to the complexity of the BM3D algorithm, which involves multiple steps like block matching and 3D filtering, leading to high computational resource and time consumption.

The DWRA method and its variant DWRA w/o SA have processing times of 0.2073 and 0.1964 seconds per image, respectively. Although slightly slower than traditional methods, these times are still acceptable. Given that watermark removal attacks are not time-sensitive and attackers typically have ample time to process images, the speed disadvantage of the DWRA method does not significantly impact its practical use. Moreover, the superior attack performance of the DWRA method, such as its ability to effectively remove various watermarks while maintaining good image quality, makes it highly valuable in practical applications.

V. CONCLUSION

The paper presents a novel Diffusion-Based Watermark Removal Attack (DWRA) that effectively removes digital watermarks from images while preserving quality. By combining latent space noise disruption with a self-attention diffusion model, DWRA operates in a black-box setting without needing watermark embedding details. Experiments show DWRA outperforms existing methods in removal efficacy and image quality preservation, highlighting watermarking system vulnerabilities and guiding the development of more robust solutions.

REFERENCES

[1] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with clip latents," *arXiv preprint arXiv:2204.06125*, 2022.

[2] C. Saharia, W. Chan, S. Saxena, L. Li, J. Whang, E. L. Denton, K. Ghasemipour, R. G. Lopes, B. K. Ayan, T. Salimans *et al.*, "Photorealistic text-to-image diffusion models with deep language understanding," in *Proceedings of the Neural Information Processing Systems (NeurIPS)*, 2022, pp. 36 479–36 494.

[3] K. Navas, M. C. Ajay, M. Lekshmi, T. S. Archana, and M. Sasikumar, "Dwt-dct-svd based watermarking," in *Proceedings of the 3rd International Conference on Communication Systems Software and Middleware and Workshops (COMSWARE'08)*. IEEE, 2008, pp. 271–274.

[4] K. A. Zhang, L. Xu, A. Cuesta-Infante, and K. Veeramachaneni, "Robust invisible video watermarking with attention," *arXiv*, vol. abs/1909.01285, 2019.

[5] S. Bezryadin, P. Bourov, and D. Ilinih, "Brightness calculation in digital image processing," in *Proceedings of the International Symposium on Technologies for Digital Photo Fulfillment (ISTDPF)*. Society for Imaging Science and Technology, 2007, pp. 10–15.

[6] C. Zhao and H. Li, "Blurring fools the network-adversarial attacks by feature peak suppression and gaussian blurring," *arXiv preprint arXiv:2012.11442*, 2020.

[7] M. Saberi, V. S. Sadasivan, K. Rezaei, A. Kumar, A. M. Chegini, W. Wang, and S. Feizi, "Robustness of ai-image detectors: Fundamental limits and practical attacks," *arXiv*, vol. abs/2310.00076, 2023.

[8] N. Lukas, A. Diaa, L. Fenaux, and F. Kerschbaum, "Leveraging optimization for adaptive attacks on image watermarks," *arXiv*, vol. abs/2309.16952, 2023.

[9] Q. Zeng, B. Liu, Y. Xu *et al.*, "An attention-based deep learning model for predicting microvascular invasion of hepatocellular carcinoma using an intra-voxel incoherent motion model of diffusion-weighted magnetic resonance imaging," *Physics in Medicine & Biology*, vol. 66, no. 18, p. 185019, 2021.

[10] B. Liu, Q. Zeng, J. Huang *et al.*, "Ivmm using convolutional neural networks predicts microvascular invasion in hcc," *European Radiology*, vol. 32, no. 10, pp. 7185–7195, 2022.

[11] Z. Wang, Q. Zeng, W. Lin *et al.*, "Generating diagnostic and actionable explanations for fair graph neural networks," in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 38, no. 19, 2024, pp. 21 690–21 698.

[12] Z. Wang, Q. Zeng *et al.*, "Multiview subgraph neural networks: Self-supervised learning with scarce labeled data," *IEEE Transactions on Neural Networks and Learning Systems*, 2024.

[13] Q. Zeng, Z. Wang, Y. Cheung *et al.*, "Ask, attend, attack: An effective decision-based black-box targeted attack for image-to-text models," in *Proceedings of the Neural Information Processing Systems (NeurIPS)*, 2024.

[14] Y. Gong, L. Huang, and L. Chen, "Eliminate deviation with deviation for data augmentation and a general multi-modal data learning method," *arXiv preprint arXiv:2101.08533*, 2021.

[15] J. Lin, Z. Wang, S. Jiang, Y. Hou, and M. Jiang, "Phys4dgen: A physics-driven framework for controllable and efficient 4d content generation from a single image," *arXiv preprint arXiv:2411.16800*, 2024.

[16] Y. Gong, Q. Zeng, D. Xu *et al.*, "Cross-modality attack boosted by gradient-evolutionary multiiform optimization," *arXiv preprint arXiv:2409.17977*, 2024.

[17] Q. Zeng, Y. Gong, and M. Jiang, "Cross-task attack: A self-supervision generative framework based on attention shift," in *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024, pp. 1–8.

[18] P. Fernandez, A. Sablayrolles, T. Furon, H. Jégou, and M. Douze, "Watermarking images in self-supervised latent spaces," in *Proceedings of International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2022, pp. 3054–3058.

[19] P. Fernandez, G. Couairon, H. Jégou, M. Douze, and T. Furon, "The stable signature: Rooting watermarks in latent diffusion models," *arXiv preprint arXiv:2303.15435*, 2023.

[20] G. K. Dziugaite, Z. Ghahramani, and D. M. Roy, "A study of the effect of jpg compression on adversarial images," *arXiv preprint arXiv:1608.00853*, 2016.

[21] E. S. Gedraite and M. Hadad, "Investigation on the effect of a gaussian blur in image filtering and segmentation," in *Proceedings of the International Symposium ELMAR*. IEEE, 2011, pp. 393–396.

[22] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-d transform-domain collaborative filtering," *IEEE Transactions on Image Processing*, vol. 16, no. 8, pp. 2080–2095, 2007.